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## 4.4 COLLISIONS

Collision is a common phenomenon that happens around us every now and then. For example, carom, billiards, marbles, etc.,. Collisions can happen between two bodies with or without physical contacts.

Linear momentum is conserved in all collision processes. When two bodies collide, the mutual impulsive forces acting between them during the collision time ($\Delta t$) produces a change in their respective momenta. That is, the first body exerts a force $\vec{F}_{12}$ on the second body. From Newton's third law, the second body exerts a force $\vec{F}_{21}$ on the first body. This causes a change in momentum $\Delta\vec{p}_1$ and $\Delta\vec{p}_2$ of the first body and second body respectively. Now, the relations could be written as,

$$\Delta\vec{p}_1 = \vec{F}_{12}\Delta t \qquad (4.44)$$

$$\Delta\vec{p}_2 = \vec{F}_{21}\Delta t \qquad (4.45)$$

Adding equation (4.44) and equation (4.45), we get

$$\Delta\vec{p}_1 + \Delta\vec{p}_2 = \vec{F}_{12}\Delta t + \vec{F}_{21}\Delta t = \left(\vec{F}_{12} + \vec{F}_{21}\right)\Delta t$$

According to Newton's third law, $\vec{F}_{12} = -\vec{F}_{21}$

$$\Delta\vec{p}_1 + \Delta\vec{p}_2 = 0$$

$$\Delta\left(\vec{p}_1 + \vec{p}_2\right) = 0$$

Dividing both sides by $\Delta t$ and taking limit $\Delta t \to 0$, we get

$$\lim_{\Delta t \to 0} \frac{\Delta\left(\vec{p}_1 + \vec{p}_2\right)}{\Delta t} = \frac{d\left(\vec{p}_1 + \vec{p}_2\right)}{dt} = 0$$

The above expression implies that the total linear momentum is a conserved quantity.
**Note:** The momentum is a vector quantity. Hence, vector addition has to be followed to find the total momentum of the individual bodies in collision.

### 4.4.1 Types of Collisions

In any collision process, the total linear momentum and total energy are always conserved whereas the total kinetic energy need not be conserved always. Some part of the initial kinetic energy is transformed to other forms of energy. This is because, the impact of collisions and deformation occurring due to collisions may in general, produce heat, sound, light etc. By taking these effects into account, we classify the types of collisions as follows:

(a) Elastic collision

(b) Inelastic collision

**(a) Elastic collision**

In a collision, the total initial kinetic energy of the bodies (before collision) is equal to the total final kinetic energy of the bodies (after collision) then, it is called as elastic collision. i.e.,

Total kinetic energy before collision = Total kinetic energy after collision

**(b) Inelastic collision**

In a collision, the total initial kinetic energy of the bodies (before collision) is not equal